

DIPLOMA – FIRST SEMESTER
MECHANICAL SCIENCE AND ENGG
UNIT – 1: ENGINEERING MATERIALS AND THEIR
PROPERTIES

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1. Write the physical properties of metals and mechanical properties of metals.
 2. Explain cast iron and types of cast iron and its applications.
 3. Define alloy cast iron.
 4. Write the effect of impurities on cast iron.
 5. Define alloy steel and effect of impurities on steel.
- Abhishek Gp Dumka**
6. What is the purpose of heat treatment? Explain briefly annealing, normalizing and hardening.
 7. Explain any four of the following: Annealing, Normalizing, Hardening, Tempering, Flame hardening, Nitriding, Cyaniding, Carburizing.
 8. What is non-metallic material?
 9. Briefly explain: Rubber, Glass, Ceramics, Polymers, Composite.
 10. What is composite and what are the different categories of composite materials? Describe the various engineering applications of these composite materials.
 11. Define stainless steel and types of stainless steel.

UNIT – 2: SHAFT, KEY AND COUPLINGS AND BEARINGS

Qno – 1 Describe the different types of shaft couplings and explain their respective applications in mechanical system.

Qno – 2 Explain sunk keys and types of sunk keys.

Qno – 3 Explain saddle keys and construction of saddle keys and write its working principle with neat sketch.

Qno – 4 Explain tangent keys and construction of tangent keys and write its working principle with neat sketch.

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Qno – 5 Explain Woodruff keys and splines – construction and working of each type with sketch.

Qno – 6 Explain sleeve or muff coupling – construction and working with sketch.

Qno – 7 Flange coupling – construction and working with sketch.

Qno – 8 Explain sliding contact bearing. Types of sliding contact bearings with advantages and disadvantages.

Qno – 9 Explain roller bearings – construction and working.

Qno – 10 Explain thrust bearings – construction and working.

Qno – 11 Explain ball bearings – construction and working.

UNIT – 3

BELT DRIVES, CHAIN DRIVES AND GEAR DRIVES

Questions:

Q1. Define belt drives and explain their importance in mechanical power transmission.

Q2. Explain the lap joints with neat sketches.

Q3. What is the difference between slip and creep in belt drive?

Q4. Write short notes on:

- a) Open belt drive
- b) Crossed belt drive
- c) Twist belt drive
- d) Compound belt drive
- e) Flat belt drive

Q5. Explain V-belt drive with sketch. Write its advantages and disadvantages.

Q6. Explain chain drives. Write its advantages and disadvantages.

Q7. Explain velocity ratio of a belt drive and write its formula.

Q8. Explain gear drive. State its advantages, disadvantages and applications.



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Q9. Explain the classification of gears with neat sketch.

Types of gears:

- a) Spur gears
- b) Helical gears
- c) Rack and pinion gears
- d) Worm gears and worm wheel

Q10. Define the velocity ratio in gear drives and explain how it affects gear performance.

Q11. Define gear train. Explain with sketches the various types of gear train.

UNIT – 4: FASTENERS

1. Explain fasteners. What is temporary and permanent fasteners? Provide examples of each.
2. Explain screw thread terminology.
3. Describe three types of screw fastenings and their applications.
4. What are locking devices and why are they important in fasteners?
5. Describe the function of any three types of screw heads with neat sketch.
6. Differentiate between temporary and permanent fasteners.
7. Define riveted joint. How does it differ from welded joint?
8. Explain the types of riveted joints with neat sketch.
9. Differentiate between lap and butt joints.

UNIT – 5: ENGINE TERMINOLOGIES

1. Explain IC engine terminologies like bore, stroke, and displacement.
2. Explain compression ratio and its effects on engine performance.
3. Define the following terms used in engine technology: TDC, BDC.
4. Define swept volume, clearance volume, and compression ratio.
5. Explain mean effective pressure (MEP).
6. Describe the following with respect to IC engine:
 - Indicated power
 - Brake power
 - Frictional power
7. Explain engine torque and its relation with engine speed and power.
8. What is specific fuel consumption (SFC)? How is it calculated?

UNIT – 6: IC ENGINES

1. Explain two-stroke IC engine construction and working.
2. Explain four-stroke SI engine construction and working.
3. Differentiate between SI and CI engines.
4. Differentiate between two-stroke and four-stroke engines.
5. Write a short note on:
 - Brake thermal efficiency
 - Indicated thermal efficiency
 - Mechanical efficiency